Designing an N-of-1 Trial from First Principles: Matching the Schedule to the Drug and the Patients, and How the Hendrickson Hybrid Compares with a Crossover

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Audience. Undergraduate statistics students who have taken a course in linear regression and know what variance, correlation, power and Type I error are. No clinical background is assumed. The companion tutorial `docs/67` builds the *analysis* of these trials. This one builds the *design*. The two can be read in either order.

Purpose. A trial design is not chosen from a catalog. It is derived from three sets of facts: what the drug does and how fast, who the patients are and what they will accept, and how the outcome behaves statistically over time. This tutorial shows how each fact turns into a design decision. It then uses the trial that motivated this project, prazosin for posttraumatic stress disorder (PTSD), to show how those decisions produce the Hybrid design of Hendrickson et al. (2020). Section 9 compares the Hybrid design with a standard one-cycle crossover and states where each is better.

Evidence. Every number below comes from an earlier project document, which is cited beside it. Most come from `docs/39`, which evaluates designs with an exact (closed-form) power formula and checks it by simulation, and from paper 01, which simulates the project's full data-generating process. Section 11 lists what the evidence does not cover.

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Notation

Symbols follow `analysis/report/NOTATION.md`. The last column gives the section where each symbol first appears.

Table 1: Notation

Symbol	Meaning	Section
N	total number of participants	2.1
i, t	participant and visit	2.1
w_t	calendar week of visit t	2.1
Y_{it}	symptom score; lower is better	2.1
B_i	biomarker measured once before treatment	2.1
D_{it}	drug state: 1 on drug, 0 off drug	2.1
$\beta_{bm:D}$	biomarker-by-treatment interaction, the target of the trial	2.1
Δ_i	participant's on-drug mean minus off-drug mean	2.2
a_t	contrast weights: $1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug	6.1
n_{on}, n_{off}	on-drug and off-drug visits per participant	6.1
t_{od}	time on drug since the last start, weeks	4.1
t_{sd}	time since the drug was stopped, weeks	4.2
$t_{1/2}$	response half-life: time for the remaining drug effect to halve	4.2
η_t	design expectancy: 1 open label, 0.5 blinded	5.1
ρ_{cs}	compound-symmetry correlation, the same for every pair of visits	6.2
ρ	AR(1) correlation per week, $\rho^{\lvert w_t - w_s \rvert}$	6.2
A	matrix of AR(1) correlations between visits	6.2
$a^\top A a$	time part of the contrast variance under AR(1)	6.2
P	number of randomization paths	6.5
c_{bm}	biomarker-drug-response correlation; the interaction strength in the extended DGP	9.1
MCSE	Monte Carlo standard error of a simulated rate, $\sqrt{\hat{\pi}(1-\hat{\pi})/R}$ for R replicates	9.1

Glossary

Terms are listed alphabetically. The number in brackets is the section where the term is introduced.

- **Aggregated N-of-1 trial.** Many single-patient on/off trials, analyzed together [2.2].
- **AR(1) correlation.** Correlation between two visits that falls geometrically with the time between them, $\rho^{\lvert w_t - w_s \rvert}$ [6.2].
- **Balance.** Equal numbers of on-drug and off-drug visits per participant [6.1].
- **Blinded.** The participant does not know whether they are taking the drug or a placebo [5.1].
- **Blinded discontinuation.** A phase in which every participant starts on drug and is switched to placebo, without being told when [7.2].
- **Carryover.** Drug effect that persists after the drug is stopped [4.2].
- **Compound symmetry (CS).** The same correlation between every pair of visits, regardless of the time between them [6.2].
- **Contrast.** A weighted sum of a participant's scores whose weights sum to zero, such as the on-minus-off difference Δ_i [2.2].
- **Counterbalancing.** Arranging sequences so that each treatment appears equally often in each period [6.5].
- **Crossover (one-cycle, AB/BA).** Two periods, drug and placebo, in randomized order [3].
- **Expectancy.** The participant's belief about being on drug. In the project's model it scales the placebo response [5.1].

- **Interaction.** Here, the change in drug effect per unit of biomarker [2.1].
- **Interleaving.** Placing on-drug and off-drug visits close together in time [6.2].
- **Measurement error.** Visit-to-visit noise in a rating that is not shared with neighboring visits [6.2].
- **Natural history.** Change in symptoms over time that would occur without treatment [5.3].
- **Onset.** How quickly the drug effect builds after starting [4.1].
- **Open label.** The participant knows they are taking the drug [5.1].
- **Path (sequence).** One planned order of drug states through the trial. Participants are randomized to paths [6.5].
- **Period effect.** A systematic difference between trial periods unrelated to treatment, such as natural improvement [3].
- **Placebo response.** Improvement caused by the belief in treatment rather than by the drug [5.1].
- **Power.** The probability that the trial detects an interaction that is present [2.1].
- **Rebound.** Worsening beyond baseline after a drug is stopped [4.4].
- **Response half-life.** The time for the remaining drug *effect* to fall by half after the drug is stopped. It can differ from the drug's half-life in blood [4.2].
- **Switch.** A change from on drug to off drug, or back, between consecutive visits [6.3].
- **Titration.** Raising the dose in steps to the target dose [4.1].
- **Washout.** An unmeasured gap after stopping a drug, so that the next measured period starts free of carryover [4.2].

1. What a design is

A design for this kind of trial is a set of answers to five questions.

1. **Which drug states, in which order?** The on/off sequence for each group of participants.
2. **Who knows?** Which phases are open label and which are blinded.
3. **When are the measurements?** The visit weeks.
4. **What is randomized?** The number of paths and how participants are allocated to them.
5. **How long, and how many visits?** Total duration and burden.

A good design answers them so that the trial estimates the quantity of interest precisely, without bias, with participants who will enroll and stay. Precision, bias and feasibility often pull in different directions. A design is a compromise among them, and the compromise depends on facts about the drug and the patients. Sections 4 to 6 collect those facts. Section 7 turns them into a design.

2. The question decides the design family

2.1 The estimand

The trial asks whether a biomarker B_i, measured once before treatment, predicts how much a participant benefits from the drug. In regression terms it asks whether the interaction $\beta_{bm:D}$ in

$$Y_{it} = \beta_0 + \beta_B B_i + \beta_T w_t + \beta_D D_{it} + \beta_{bm:D} B_i D_{it} + ...$$

is different from zero. The design's job is to make that coefficient estimable with as little variance as possible.

2.2 Why each participant must be seen on and off drug

In a parallel-group trial each participant is on drug or on placebo, never both. The drug effect of one person is never observed. The interaction must be estimated by comparing the biomarker slope in the drug arm with

the slope in the placebo arm. Two slopes from two groups of different people carry all the between-person variation in symptom level.

If instead every participant is observed both ways, each has their own on-minus-off contrast

$$\Delta_i = \overline{Y}_{i,\text{on}} - \overline{Y}_{i,\text{off}}.$$

The person's own baseline severity cancels out of Δ_i. The interaction is then the slope of Δ_i on B_i. Because the largest source of noise has been removed, far fewer participants are needed. This is the rationale for an **aggregated N-of-1 design**. An interaction test needs about four times the sample of a main-effect test of the same size (Brookes et al., 2004), so this saving matters more for interactions than for ordinary treatment effects.

Design consequence. Every participant should spend time both on and off drug. The remaining questions are how much, in what order, and when to measure.

3. The baseline candidate: the one-cycle crossover

The simplest design in which every participant is seen both ways is the **two-period crossover**, often written AB/BA. Half the participants receive drug then placebo; the other half receive placebo then drug. The project's version, taken from Hendrickson et al. (2020), has two 10-week blocks, eight visits every 2.5 weeks, blinding throughout and no washout (`docs/series/ch02`).

Table 2: The one-cycle crossover (CO)

Path	Drug state at weeks 2.5, 5, …, 20
A	1 1 1 1 0 0 0 0
B	0 0 0 0 1 1 1 1

Its strengths are well known (Senn, 2002).

- **Balance.** Four on-drug and four off-drug visits per participant.
- **Constant blinding.** Expectancy is the same at every visit, so the on/off contrast is not mixed up with a change in what the participant believes.
- **Period effects are separable.** Because the two sequences are mirror images, a systematic change between the first and second half of the trial can be estimated and removed.
- **A carryover-free path.** In the placebo-first path, every off-drug visit comes before any drug exposure.

Its weaknesses are also well known.

- **Carryover in the drug-first path.** The first placebo visits follow the drug directly. Classical analyses of the two-period crossover struggle to separate carryover from the treatment effect (Grizzle, 1965; Freeman, 1989). A washout gap is the usual remedy.
- **Half the participants start on placebo.** Some patients will not enroll on those terms (Section 5.2).
- **One switch per participant.** Each participant's on and off visits are far apart in time. Section 6.2 shows why that matters when correlation fades with time.

The rest of this tutorial asks how each property of the drug and the population should change this starting point.

4. Properties of the drug

4.1 Onset: how fast the effect builds

Some drugs act within hours. Others take weeks. Prazosin must be titrated, and Hendrickson et al. assumed about 2.5 weeks of titration each time it is started. The project's model of the drug response follows a slow-start S-shaped (Gompertz) curve in time on drug, t_{od}. At the published parameters the response reaches 3 percent of its ceiling after one week on drug, 39 percent after four weeks and 84 percent after eight (`docs/40`, Section 4).

Design consequences.

- An on-drug period must be long enough for the effect to appear. A one-week on-drug period would measure almost nothing.
- Each restart costs time. In the model, time on drug resets at every restart, so a four-week re-exposure recovers only 39 percent of the ceiling. (This reset is a modeling choice. `docs/38` argues the response may rebuild from its residual level instead.)
- Slow onset therefore favors **few, long on-drug periods**, and it argues for putting a long on-drug phase early, where it also serves as titration.

4.2 Offset: carryover and the response half-life

When the drug stops, its effect decays. Write $t_{1/2}$ for the **response half-life**: the time for the remaining effect on symptoms to halve. It is not the same as the drug's half-life in blood. A drug can leave the blood in hours while its effect on symptoms lasts longer (`docs/38`). One half-life after stopping, half the effect remains. Two half-lives after, a quarter.

An off-drug visit taken before the effect has decayed is contaminated. It is coded "off" but still carries drug effect. This shrinks the measured on-minus-off contrast and costs power.

Design consequences.

- Off-drug visits should fall several half-lives after a discontinuation. `docs/39` (Section 5) finds the best spacing between visits to be about two to six half-lives, depending on how noisy the measurement is.
- Each switch from on to off creates one contaminated window. A long half-life therefore favors **fewer switches**. With a two-week half-life under compound symmetry, the best one-switch design has power 0.731 and the best seven-switch design 0.559 (`docs/39`, Section 4).
- The alternatives are a **washout**, an unmeasured gap that lengthens the trial, or an analysis that models the decay. The second option is the subject of `docs/67`, Section 8.

4.3 Reversibility

An N-of-1 design assumes that the condition returns toward its untreated state when the drug stops. A drug that cures, or that changes the course of the disease permanently, cannot be studied this way. Every off-drug period after the first exposure would still show the benefit. Reversibility is a precondition, not a design choice.

4.4 Safety of stopping

Some drugs cause rebound or withdrawal symptoms when stopped abruptly. A design that stops the drug must then stop it in a controlled way and monitor closely afterwards. Frequent visits right after a discontinuation serve both safety and measurement.

5. Properties of the population

5.1 Expectancy and the placebo response

Patients who know they are on drug improve more than patients who are unsure. The project models this as a placebo-response component scaled by the design expectancy η_t: 1 when the participant knows they are on drug (open label), 0.5 when blinded (`docs/series/ch02`).

If the open-label visits are all on drug and the blinded visits are a mixture, part of the on-minus-off contrast is really an open-label-versus-blinded contrast. The design has mixed the drug effect with an expectancy effect. In the Hybrid design the correlation between the drug indicator and the expectancy variable is 0.59 (`docs/40`, Section 1). The analysis must then model expectancy, and any error in that model adds noise to the contrast.

Design consequence. Keep expectancy the same at the on-drug and off-drug visits that form the contrast. Blind the contrast.

5.2 Willingness to enroll

Patients who are acutely ill may refuse a trial in which they could start on placebo. Hendrickson et al. included an open-label phase so that such patients could enroll; they reported that recruitment then outpaced the budget (`docs/40`, Section 2). A trial that cannot recruit has no power at all, so this consideration can outweigh any statistical one.

Design consequence. For a symptomatic population, start everyone on drug, open label. Section 5.1 says this has a statistical cost. `docs/39` (Section 7) measures it. With eight visits in total, each open-label visit that *replaces* a blinded visit lowers power. Two open-label visits cost 0.09 to 0.15 of power under compound symmetry and 0.09 to 0.13 under AR(1) with 20 percent measurement error. The same document notes a cheaper option that was not evaluated: *add* the open-label phase before the blinded visits rather than in place of them, and leave it out of the on/off contrast.

5.3 Natural history

Many conditions improve or worsen on their own over months. If participants are on drug early and off drug late, a natural improvement over time looks like a drug effect, or hides one. The analysis adjusts for time, but only as well as its model of time is correct.

Design consequence. Randomize *when* the drug stops, so that at some visits some participants are on drug and others are off at the same calendar time. A comparison between paths at the same visit is protected by randomization, not by a model of time (`docs/40`, Section 7).

5.4 Dropout and burden

Longer trials and more visits lose more participants. If participants drop out because they feel worse off drug, dropout depends on treatment, and the missing visits are not missing at random. Paper 09 studies this in detail.

Design consequences. Keep the trial no longer than the onset and offset require. Collect informative on-drug visits early. Do not make off-drug periods longer than the carryover requires.

6. Statistical properties of the outcome

6.1 Balance

The variance of Δ_i contains a term proportional to $1/n_{\text{on}} + 1/n_{\text{off}}$. With eight visits, that sum is 0.50 for four and four, 0.53 for five and three, and 0.67 for six and two. The balance term is therefore 7 percent larger at five and three and 33 percent larger at six and two (`docs/39`, Section 3).

Design consequence. Aim for equal numbers of on-drug and off-drug visits in the contrast.

6.2 How correlation behaves over time

The contrast $\Delta_i = \sum_t a_t Y_{it}$, with weights $a_t = 1/n_{\text{on}}$ on drug and $-1/n_{\text{off}}$ off drug, has variance $a^\top C a$, where C is the covariance of a participant's eight scores. What the design can do about this variance depends on the shape of C. Two shapes bracket the possibilities.

Compound symmetry (CS). Every pair of visits has the same correlation ρ_{cs}, whether one week or sixteen weeks apart. Then the order of the on and off visits is irrelevant. Only the balance and the blinding matter. Every blinded four-and-four design has the same contrast variance (`docs/39`, Section 3).

AR(1). Correlation fades with the time between visits, $\rho^{|w_t - w_s|}$. A participant's slowly drifting "current state" is shared by nearby visits and cancels from a difference of nearby visits. It does not cancel from a difference of distant visits. Under AR(1) the variance factors into a part from the components and a time part $a^\top A a$, where $A_{ts} = \rho^{|w_t - w_s|}$. Only the time part depends on the arrangement of visits.

The following R code computes the time part for several designs at $\rho = 0.8$ per week.

```r
time_part <- function(weeks, on, rho = 0.8) {
  a <- ifelse(on == 1, 1 / sum(on), -1 / sum(1 - on))
  A <- rho^abs(outer(weeks, weeks, '-'))
  drop(t(a) %*% A %*% a)
}
co <- seq(2.5, 20, by = 2.5)
hybrid <- c(4, 8, 9, 10, 11, 12, 16, 20)
time_part(co, c(1, 1, 1, 1, 0, 0, 0, 0))      # crossover
time_part(hybrid, c(1, 1, 1, 0, 0, 0, 1, 0))  # Hybrid, path 3
time_part(co, c(1, 0, 1, 0, 1, 0, 1, 0))      # alternating
```

Table 3: Time part of the contrast variance under AR(1), $\rho = 0.8$. The crossover, Hybrid and alternating values were recomputed with the code above for this tutorial; all agree with `docs/39`, Section 3.

Design	Switches per path	$a^\top A a$
Crossover (CO)	1	0.828
Open label + blinded discontinuation	1	0.649 to 0.664
Hybrid	2 to 3	0.361 to 0.449
Alternating every visit, every 2.5 weeks	7	0.165

The alternating design has about a fifth of the crossover's time part. The Hybrid sits between them, because its weekly blinded-discontinuation visits put off-drug visits right next to on-drug ones.

Design consequence. If correlation fades with time, interleave on and off visits. If it does not, interleaving buys nothing and only adds switches, which carryover penalizes (Section 4.2). Which shape real symptom data follow is an empirical question. Pilot data or earlier trials should be used to judge it.

6.3 Switches against carryover

Sections 4.2 and 6.2 pull in opposite directions. More switches put on and off visits closer together, which helps under AR(1). Each switch also creates a contaminated off-drug window, which hurts when the half-life is long. `docs/39` (Section 4) gives the best power attainable with a given number of switches. Under AR(1) with 20 percent measurement error and a one-week half-life, it rises from 0.221 with one switch to 0.435 with seven. Under compound symmetry with a two-week half-life, it falls from 0.731 to 0.559.

6.4 Measurement error

Rating scales add noise that is not shared with the next visit. Under pure AR(1), visits placed very close together would be almost perfectly correlated, and the contrast noise would vanish. Measurement error puts a floor under that. The more of the variance is measurement error, the less close spacing helps, and the further apart the visits should be (`docs/39`, Section 5).

6.5 Randomization and paths

Randomizing participants to two or more paths does two things.

- It allows **between-path comparisons at the same visit**, which identify the drug effect without relying on a model of time or expectancy (Section 5.3).
- It allows **counterbalancing**, so that period effects can be estimated and removed.

For the interaction specifically, the mixture of paths matters less than one might expect. In the Hybrid design, putting all 70 participants on one path changed interaction power by at most 0.04 (`docs/40`, Section 7). The interaction is estimated from each participant's own contrast. The randomization protects against a biomarker that is related to the natural history or to expectancy.

7. Worked example: deriving a design for prazosin in PTSD

This section applies Sections 4 to 6 to the trial that motivated the project (NCT03539614). The question was whether standing systolic blood pressure at baseline predicts the reduction in PTSD symptoms that prazosin produces (`docs/40`, Section 2).

7.1 The facts

Table 4: Inputs for the prazosin design

Fact	Value	Source
Onset	slow; titration about 2.5 weeks; 84 percent of effect by 8 weeks	4.1
Offset	plausibly short pharmacologically (0.1 to 0.2 week); longer behavioral persistence possible	4.2
Reversibility	symptomatic treatment; benefit expected to fade off drug	4.3
Population	acutely symptomatic; reluctance to accept placebo at entry	5.2
Natural history	symptoms may improve over months	5.3

Fact	Value	Source
Correlation	unknown; plausibly fades with time	6.2

7.2 The decisions

1. **Start everyone on drug, open label, for eight weeks.** This answers the recruitment problem (5.2). It doubles as titration (4.1). Eight weeks is long enough for the slow-onset response to reach most of its ceiling. Visits at weeks 4 and 8 record the on-drug response. The cost is expectancy mixing (5.1), which the analysis must model.
2. **Then blind, while everyone is still on drug.** From week 8, participants are told they may be switched to placebo. Everyone remains on drug at week 9, so the drop in expectancy can be seen without a change in drug.
3. **Randomize when the drug stops.** Half the participants stop after week 9, half after week 10. At week 10 the two groups differ only in drug state, at the same expectancy and calendar time. This is a randomized comparison (5.3, 6.5). Weekly visits at weeks 9 to 12 place off-drug visits next to on-drug ones, which helps if correlation fades with time (6.2). They also monitor safety after stopping (4.4).
4. **End with a short blinded crossover.** Two four-week periods, drug and placebo in randomized order, with one visit at the end of each (weeks 16 and 20). This re-exposure adds a second randomized on/off comparison at the same calendar time. Four weeks also lets a short carryover decay before the off-drug visit (4.2).
5. **Two randomizations make four paths.** Two stop times crossed with two crossover orders.

The result is the Hybrid design (Table 5, from `docs/40`, Section 2).

Table 5: The Hybrid design. Weeks 4 and 8 are open label; weeks 9 to 20 are blinded.

Path	Drug state at weeks 4, 8, 9, 10, 11, 12, 16, 20	Weeks since stopping
1	1 1 1 1 0 0 1 0	0 0 0 0 1 2 0 4
2	1 1 1 1 0 0 0 1	0 0 0 0 1 2 6 0
3	1 1 1 0 0 0 1 0	0 0 0 1 2 3 0 4
4	1 1 1 0 0 0 0 1	0 0 0 1 2 3 7 0

7.3 Different facts, different designs

The same reasoning gives other designs for other drugs and populations. These are illustrations of the principles, not evaluated recommendations.

- **Fast onset, short carryover, a stable chronic condition, patients who accept placebo.** No open label is needed. If correlation fades with time, a blinded design that alternates every two to four weeks is close to the best design in every condition `docs/39` examined (Section 6 there).
- **Long carryover, compound-symmetry-like correlation.** Few, long blocks: a blinded crossover with wide spacing, or with a washout before the off-drug visits.
- **Slow onset and patients who accept placebo.** A long blinded on-drug period first, then a blinded discontinuation, without the open-label phase. Under compound symmetry, running the Hybrid schedule fully blinded raised its power from 0.663 to 0.739 (`docs/39`, Section 3, decayed mean moderation, no carryover).
- **Patients who refuse placebo at entry, and a need for maximum power.** Add the open-label phase *before* the eight blinded visits instead of replacing two of them, and leave it out of the contrast (`docs/39`, Section 7, not evaluated there).

8. A design procedure

The steps below summarize Sections 4 to 7. They follow the procedure in `docs/39`, Section 10.

1. **State the estimand.** Here, the biomarker-by-treatment interaction.
2. **Check reversibility.** If the effect does not fade off drug, do not use an N-of-1 design.
3. **Estimate onset and the response half-life,** from pharmacology, earlier trials or pilot data. Give a plausible range, not a point.
4. **Judge the correlation structure and the measurement-error share,** from earlier repeated-measures data on the same outcome.
5. **List the population constraints:** acceptance of placebo at entry, tolerance of withdrawal, expected dropout, maximum length.
6. **Blind and balance the contrast.** Equal on and off visits at equal expectancy. If an open-label phase is required, add it before the blinded visits rather than in place of them.
7. **Choose the number of switches and the spacing** against the half-life and the correlation structure (Sections 4.2, 6.2, 6.3).
8. **Randomize the timing of at least one switch,** so that some visits compare paths at the same calendar time.
9. **Evaluate candidates by simulation or closed form across the plausible ranges,** and prefer a design that stays near the best in all of them over one that is best in only one.
10. **Fix the analysis together with the design.** The analysis of `docs/67` and the design interact: for example, estimating the half-life from the data needs more than one switch per path (Section 9).

9. The Hybrid design against the one-cycle crossover

Both designs have eight visits over 20 weeks. Table 6 compares them on each criterion of Sections 4 to 6.

Table 6: Hybrid and one-cycle crossover compared

Criterion	Hybrid	Crossover	Better
Recruitment of symptomatic patients	everyone starts on drug, open label	half start on placebo	Hybrid
Titration	built into the open-label phase, before any blinded visit	falls inside a blinded measured period, at week 10 for the placebo-first path	Hybrid
Blinding of the contrast	open-label visits are all on drug; expectancy mixed with drug state	constant expectancy	Crossover
Balance of on and off visits	5 and 3 (paths 1, 2) or 4 and 4 (paths 3, 4)	4 and 4	Crossover, slightly
Contrast noise under compound symmetry	larger	smaller	Crossover
Contrast noise under AR(1)	time part 0.36 to 0.45	time part 0.83	Hybrid
Robustness to carryover with a binary drug indicator	off-drug visits one to three weeks after stopping	placebo-first path free of carryover; long off-drug blocks	Crossover
Estimating the half-life from the data	two or three switches per path	one switch per path	Hybrid
Dependence on the model of time	drug state mixed with early trial time	period effect separable by design	Crossover
Randomized same-visit comparisons	weeks 10, 16, 20	every visit	Crossover

The evidence behind the rows:

- **Compound symmetry favors the crossover.** Under CS the contrast variance is 35.9 for the blinded crossover against 44.1 and 45.0 for the Hybrid paths (`docs/39`, Section 3). With decayed mean moderation and no carryover the power of the analysis studied there was 0.765 for the crossover and 0.663 for the Hybrid. Most of the difference comes from the open-label phase: the same Hybrid schedule run blinded reached 0.739. The rest comes from the five-and-three imbalance of paths 1 and 2, whose blinded contrast variance is 38.3 against 35.9 for paths 3 and 4 and the crossover. Under CS the arrangement of visits does not matter, so the Hybrid's interleaving brings no gain there.
- **AR(1) favors the Hybrid.** Its time part is about half the crossover's (Section 6.2). The gain comes from the weekly blinded-discontinuation visits and the extra switches, not from the open-label phase.
- **Under the project's full simulation the Hybrid has more power when carryover is short.** Paper 01 simulated both designs under the extended DGP, which uses AR(1) correlation, and analyzed them with G9 (`docs/67`). At $N = 70$, a moderate biomarker effect and a 0.2-week half-life, G9 power was 0.650 (MCSE 0.015) in the Hybrid design and 0.413 (MCSE 0.016) in the crossover. Paper 01 attributes part of the crossover's deficit to the analysis: the crossover has constant expectancy and only one switch per path, which gives G9 little information to choose a half-life. Paper 01 did not separate that from the correlation structure. Section 9.1 shows that the advantage disappears at a one-week half-life.
- **A simple analysis loses more to carryover in the Hybrid.** At a one-week half-life, an analysis with a binary drug indicator lost a fifth to two thirds of its power in the Hybrid and OL+BDC designs, and a tenth to a sixth in the crossover (paper 02, Introduction). The Hybrid's off-drug visits come soon after stopping. The crossover's placebo-first path has none of that contamination.
- **The crossover depends less on the model of time.** In the Hybrid design, a flexible time adjustment (a spline or visit as a factor) raised interaction power but also raised the Type I error to 0.075 and 0.080 under one-week carryover, against 0.056 for a linear time term. In the crossover all three time adjustments held their size (`docs/40`, Section 1).

9.1 Simulated size and power

Two stored simulation grids run both designs under the same extended DGP. No new simulation was run for this section. The script `analysis/scripts/tutorial-design/01-co-vs-hybrid-heatmaps.R` reads the stored summaries and writes the figures below and the tables in `analysis/data/tutorial-design/`.

- **Grid 1, paper 01 (simrun job 40).** G9 only, $N = 70$, biomarker correlation $c_{bm} \in \{0, 0.25, 0.35, 0.45\}$, carryover half-life 0, 0.2 and 1 week. There are 2,000 replicates per null cell and 1,000 per alternative cell.
- **Grid 2, paper 02 (simrun job 27).** The whole analysis ladder of `docs/67`, $N = 35$ and 70, 500 replicates per cell. The null cells have a one-week half-life. The power cells have $c_{bm} = 0.45$ and half-lives of 0, 0.5 and 1 week. With 500 replicates the MCSE of a rejection rate near 0.05 is 0.010, so rates between about 0.03 and 0.07 are consistent with a correct test.

c_{bm} is the correlation between the biomarker and the drug response, the way the extended DGP encodes the interaction. Larger values mean a stronger interaction.

Size. Table 7 gives the Type I error of G9 in grid 1.

Table 7: G9 Type I error (MCSE) by carryover half-life, $N = 70$, 2,000 replicates per cell (grid 1)

Design	0 wk	0.2 wk	1 wk
Hybrid	0.045 (0.005)	0.051 (0.005)	0.051 (0.005)
Crossover	0.059 (0.005)	0.048 (0.005)	0.049 (0.005)

Table 8 and Figure 1 give the Type I error of every analysis in grid 2.

Table 8: Type I error by analysis, one-week half-life, 500 replicates per cell, MCSE about 0.010 (grid 2)

Analysis	Hybrid, $N = 35$	Hybrid, $N = 70$	Crossover, $N = 35$	Crossover, $N = 70$
E0	0.070	0.060	0.150	0.128
G1	0.042	0.032	0.030	0.046
G6	0.066	0.046	0.046	0.054
G3	0.030	0.034	0.034	0.046
G8	0.044	0.054	0.052	0.050
G4	0.042	0.038	0.042	0.050
G9	0.060	0.058	0.058	0.056
G5 (E9)	0.052	0.056	0.054	0.038

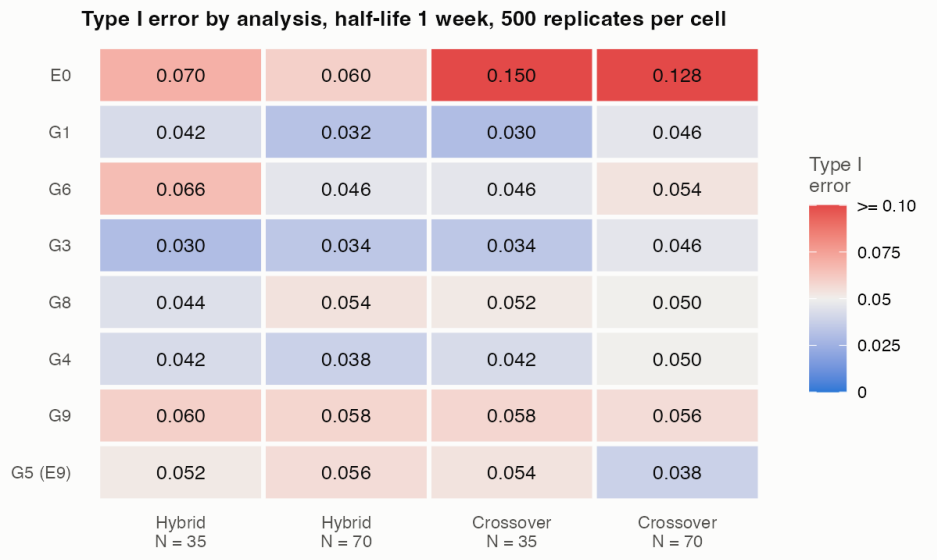


Figure 1: Type I error by analysis and design, one-week half-life, 500 replicates per cell (grid 2). Gray is the nominal 0.05; red is liberal, blue conservative. Rates of 0.10 or more share the darkest red.

Three points follow.

1. **G9 holds its size in both designs.** Its rejection rate is 0.045 to 0.059 in grid 1 and 0.056 to 0.060 in grid 2. The design does not change this.
2. **The original analysis E0 is liberal in the crossover.** It rejects a true null at 0.150 and 0.128, against 0.070 and 0.060 in the Hybrid. E0 has no residual correlation (`docs/67`, Section 6). A plausible reason for the larger excess in the crossover, not tested here, is that its on and off visits are far apart in time, where the omitted correlation changes the variance of the contrast most.
3. **The other analyses are close to nominal in both designs.** The model-based analyses G1, G3 and G4 lean conservative (0.030 to 0.050). No analysis other than E0 lies outside 0.03 to 0.07.

Power. Table 9 and Figure 2 give G9 power in grid 1.

Table 9: G9 power (MCSE) at $N = 70$, 1,000 replicates per cell (grid 1). The standard error of the difference treats the two designs as independent.

c_{bm}	Half-life	Hybrid	Crossover	Hybrid minus crossover
0.25	0 wk	0.663 (0.015)	0.351 (0.015)	+0.312 (0.021)
0.25	0.2 wk	0.650 (0.015)	0.413 (0.016)	+0.237 (0.022)
0.25	1 wk	0.418 (0.016)	0.387 (0.015)	+0.031 (0.022)
0.35	0 wk	0.919 (0.009)	0.651 (0.015)	+0.268 (0.017)
0.35	0.2 wk	0.919 (0.009)	0.626 (0.015)	+0.293 (0.018)
0.35	1 wk	0.669 (0.015)	0.647 (0.015)	+0.022 (0.021)
0.45	0 wk	0.997 (0.002)	0.874 (0.010)	+0.123 (0.011)
0.45	0.2 wk	0.994 (0.002)	0.867 (0.011)	+0.127 (0.011)
0.45	1 wk	0.854 (0.011)	0.828 (0.012)	+0.026 (0.016)

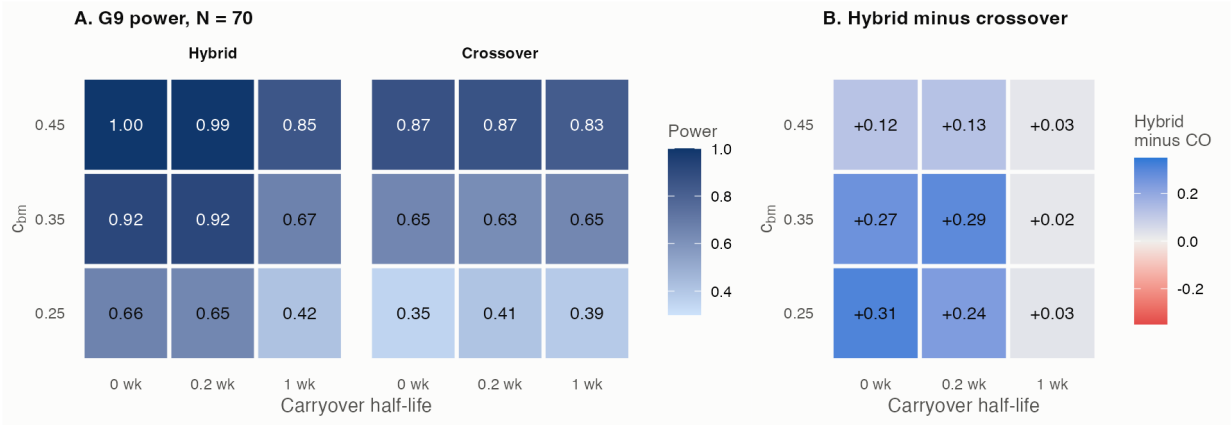


Figure 2: G9 power at $N = 70$ by biomarker correlation and carryover half-life (A), and the Hybrid-minus-crossover difference (B), grid 1. The power scale starts at 0.3. In B, blue favors the Hybrid and red the crossover.

Table 10 and Figure 3 give the power of every analysis in grid 2 at $c_{bm} = 0.45$ and $N = 70$. Figure 3 also shows $N = 35$.

Table 10: Power by analysis at $c_{bm} = 0.45$, $N = 70$, 500 replicates per cell, MCSE at most 0.022 (grid 2). E0's power is not comparable with the others in the crossover, because E0 is liberal there.

Analysis	Hybrid 0 wk	Hybrid 0.5 wk	Hybrid 1 wk	Crossover 0 wk	Crossover 0.5 wk	Crossover 1 wk
E0	0.986	0.950	0.846	0.940	0.936	0.934
G1	0.992	0.960	0.790	0.880	0.874	0.832
G6	0.996	0.972	0.831	0.878	0.874	0.846
G3	0.992	0.960	0.886	0.880	0.874	0.852
G8	0.996	0.976	0.904	0.878	0.874	0.842
G4	0.990	0.962	0.876	0.878	0.874	0.844
G9	0.996	0.974	0.900	0.880	0.880	0.850
G5 (E9)	0.986	0.950	0.848	0.788	0.780	0.752

Four points follow.

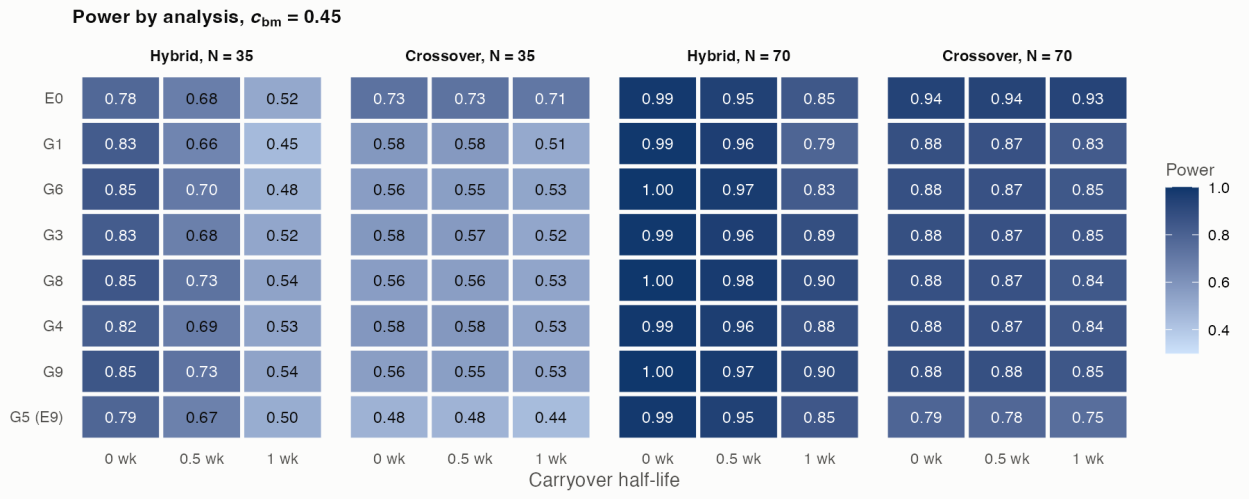


Figure 3: Power by analysis, design, sample size and carryover half-life at $c_{bm} = 0.45$ (grid 2). The power scale starts at 0.3. E0 is liberal in the crossover (Figure 1), so its power there overstates what a correctly sized test would achieve.

1. **Without carryover the Hybrid is much more powerful.** At $c_{bm} = 0.25$ G9 power is 0.663 in the Hybrid and 0.351 in the crossover, a difference of 0.31 (SE 0.02). At $N = 35$ and $c_{bm} = 0.45$ in grid 2 it is 0.85 against 0.56. This is the interleaving advantage of Section 6.2. The extended DGP uses AR(1) correlation, which rewards on and off visits close in time.
2. **At a one-week half-life the advantage disappears.** The G9 difference is +0.031, +0.022 and +0.026 at the three effect sizes, none more than two standard errors from zero. The Hybrid loses a third or more of its power between 0 and 1 week at $c_{bm} = 0.25$ (0.663 to 0.418). The crossover loses almost none (0.351 to 0.387; the rise is within Monte Carlo error). Most of the Hybrid's off-drug visits fall one to three weeks after stopping. The crossover's off-drug visits fall 2.5 to 10 weeks after stopping in one path and before any exposure in the other.
3. **The analysis matters more in the Hybrid.** At a one-week half-life and $N = 70$, the binary-indicator analysis G1 has power 0.790 in the Hybrid and the exposure-weighted G8 0.904, a gain of 0.11. In the crossover every G analysis lies between 0.832 and 0.852. Modeling carryover pays where off-drug visits are close to a discontinuation. With G1 the crossover is slightly more powerful than the Hybrid at one week (0.832 against 0.790).
4. **The simple paired difference suffers most in the crossover.** G5 (E9) has power 0.752 to 0.788 in the crossover against 0.832 to 0.880 for the mixed models. Each crossover participant contributes one on-minus-off contrast between blocks ten weeks apart, and the AR(1) noise does not cancel between them.

9.2 Verdict

Neither design dominates. The Hybrid's statistical advantage is its interleaving. That advantage exists only if correlation fades with time, which the project's main simulation assumes but which must be checked for a real outcome. In the project's simulations it is large when the response half-life is a few days and gone at one week, where the crossover's long off-drug blocks escape carryover. Its open-label phase is a statistical cost, paid for recruitment and titration. The crossover is better when correlation is roughly constant over time, when the analysis is simple, and when separating period effects cleanly matters most. If patients will accept placebo at entry and correlation fades with time, a blinded design that alternates more often than either is likely to beat both (`docs/39`, Section 6). If patients will not accept placebo at entry, the Hybrid's open-label start is the reason the trial can run at all.

10. Summary of principles

1. The estimand decides the family: an interaction needs each participant on and off drug.
2. The drug decides period lengths: slow onset needs long on-drug periods; long carryover needs few switches and wide spacing.
3. The population decides what is feasible: an open-label start recruits symptomatic patients, at a measured cost in power.
4. The correlation structure decides the arrangement: interleave if correlation fades with time; otherwise only balance and blinding matter.
5. Randomize the timing of a switch, to protect the drug-effect comparison from the model of time.
6. Evaluate designs across the plausible range of the uncertain inputs, and choose one that is robust.

11. What the evidence does not cover

- **All numbers are from models.** The closed-form results of `docs/39` assume a mean-shift form of the interaction, $N = 70$, two sequences of equal size and equal visit spacing. Paper 01 uses a correlation form of the interaction and AR(1) correlation. Neither was fitted to real PTSD symptom data.
- **The correlation structure is unknown.** The ranking of the Hybrid and the crossover reverses between compound symmetry and AR(1). This is the most important unknown for a real design.
- **The restart rule is an assumption.** The model restarts the drug response from zero at every re-exposure. If the response rebuilds from its residual level, short re-exposure periods are less costly.
- **Expectancy enters only the noise.** In the models used here the biomarker is unrelated to the placebo response. If it were related, an open-label phase could bias the interaction, not only add noise.
- **Dropout and recruitment were not modeled jointly with the design.** The recruitment benefit of the open-label phase is reported by Hendrickson et al.; it is not a simulated quantity.

12. Exercises

1. Use `time_part()` from Section 6.2 to compute $a^\top A a$ for all four Hybrid paths and for an ABAB design at weeks 2.5 to 20. Repeat at $\rho = 0.5$. Does the ranking of the designs change?
2. Show that for a blinded design under compound symmetry the contrast variance depends only on n_{on} and n_{off}, not on their order.
3. With a response half-life of one week, what fraction of the drug effect remains at each off-drug visit of each Hybrid path? Which path's off-drug visits are least contaminated?
4. A drug reaches 90 percent of its effect within three days and has a response half-life of one day. Patients accept placebo. Propose an eight-visit, eight-week design and justify each choice with a section of this tutorial.
5. Redesign the Hybrid trial so that the open-label phase is added before eight blinded visits instead of replacing two of them. How long is the trial? What is gained, and what does it cost the participants?
6. Explain why randomizing the stop time (week 9 or week 10) protects the estimate of the drug effect against natural improvement over time, and why it does less for the interaction.
7. A colleague claims that the Hybrid design is more powerful than the crossover. Write a one-paragraph reply stating under what conditions the claim holds and under what conditions it fails.

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